

FINS3616 International Business Finance Term 2, 2026

Week 2 and possibly part of Week 3 Reading Guide and Tutorial Questions

Chapters 6, and 4: Foreign Exchange Markets, Parity Conditions: PPP

Week 2 Readings (Shapiro)

Chapter 6 – The Foreign Exchange Market

6.1 Organization of the Foreign Exchange Market

6.2 The Spot Market

6.3 The Forward Market

Chapter 4 – Parity Conditions in International Finance and Currency Forecasting

4.1 Arbitrage and the Law of One Price

4.2 Purchasing Power Parity

Chapter 6:

Forward Markets: Problems 2, 7, 8 and 9

Chapter 4:

Conceptual Questions 1, 3

Absolute PPP: Problem 2

Relative PPP: Problem 7, 8

(Non-Shapiro) Additional Questions 1, 2

Chapter 6

Chapter 6, Problem 2

The \$/€ exchange rate is €1 = \$1.45, and the €/SFr exchange rate is SFr 1 = €0.71. What is the SFr/\$ exchange rate?

ANSWER. $\text{SFr}1 = €0.71 \times 1.45 = \1.0295 . $\text{USD } 1 = 1/1.0295 = \text{SFr}0.9713$

Alternatively, $\text{SFr}/\$ = \text{SFr}/€ \times €/ \$ = 1/0.71 \times 1/1.45 = 0.9713$

Chapter 6, Problem 7

Suppose Credit Suisse quotes spot and 90-day forward rates on the Swiss franc of \$0.7957-60, 8-13.

a. *What are the outright 90-day forward rates that Credit Suisse is quoting?*

ANSWER. The outright forwards are: bid rate = \$0.7965 (0.7957 + 0.0008) and ask rate = \$0.7973 (0.7960 + 0.0013).

- b. *What is the forward discount or premium associated with buying 90-day Swiss francs?*

ANSWER. The annualized forward premium = $[(0.7973 - 0.7960)/0.7960] \times 4 = 0.65\%$.

- c. *Compute the percentage bid-ask spreads on spot and forward Swiss francs.*

ANSWER. The bid-ask spread is calculated as follows:

$$\text{Percent spread} = \frac{\text{Ask price} - \text{Bid price}}{\text{Ask price}} \times 100$$

Substituting in the numbers yields a spot bid-ask spread of $(0.7960 - 0.7957)/0.7960 = 0.04\%$. The corresponding forward bid-ask spread is $(0.7973 - 0.7965)/0.7973 = 0.10\%$.

Chapter 6, Problem 8

Suppose Dupont receives quotes of \$0.009369-71 for the yen and \$0.03675-6 for the Taiwan dollar (NT\$).

- a. *How many U.S. dollars will Dupont receive from the sale of ¥50 million?*

ANSWER. Dupont must sell yen at the bid rate, meaning it will receive from this sale \$468,450 ($50,000,000 \times 0.009369$).

- b. *What is the U.S. dollar cost to Dupont of buying ¥1 billion?*

ANSWER. Dupont must buy at the ask rate, meaning it will cost Dupont \$9,371,000 ($1,000,000,000 \times 0.009371$) to buy ¥1 billion.

- c. *How many NT\$ will Dupont receive for U.S.\$500,000?*

ANSWER. Dupont must sell at the bid rate for U.S. dollars (which is the reciprocal of the ask rate for NT\$, or $1/0.03676$), meaning it will receive from this sale of U.S. dollars NT\$13,601,741 ($500,000/0.03676$).

- d. *How many yen will Dupont receive for NT\$200 million?*

ANSWER. To buy yen, Dupont must first sell the NT\$200 million for U.S. dollars at the bid rate and then use these dollars to buy yen at the ask rate. The net result from these transactions is ¥784,334,649.45 ($200,000,000 \times 0.03675/0.009371$).

- e. *What is the yen cost to Dupont of buying NT\$80 million?*

ANSWER. Dupont must sell the yen for dollars at the bid rate and then buy NT\$ at the ask rate with the U.S. dollars. The net yen cost to Dupont from carrying out these transactions is ¥313,886,220.51 ($80,000,000 \times 0.03676/0.009369$).

Chapter 6, Problem 9

Suppose the GBP/EUR exchange rate is quoted at 0.6064–0.6080 in London, and the EUR/GBP exchange rate is quoted at 1.6244–1.6259 in Frankfurt

- a. *Is there a profitable arbitrage situation? Describe it.*

ANSWER. First, convert the London GBP/EUR into a EUR/GBP quote:

$$\text{EUR/GBP: } 1/0.6080 - 1/0.6064$$

$$\text{EUR/GBP: } 1.6447 - 1.6491$$

Compare this quote with the one in Frankfurt, the arbitrage strategy is: buy GBP in Frankfurt for 1.6259 EUR then sell GBP in London for 1.6447 EUR. The profit per GBP is $1.6447 - 1.6259 = 0.0188$ EUR or $0.0188/1.6259 = 0.0116$ (1.16%)

- b. *Compute the percentage bid ask spreads on the pound and euro.*

ANSWER. The percentage bid-ask spreads on the pound and euro are calculated as follows:

$$\text{£ bid-ask spread} = (1.6259 - 1.6244)/1.6259 = 0.09\%$$

$$\text{euro bid-ask spread} = (0.6080 - 0.6064)/0.6080 = 0.26\%$$

Chapter 4

Chapter 4, Question 1

1. a. *What is purchasing power parity?*

ANSWER. In its absolute version, purchasing power parity states that price levels should be equal worldwide when expressed in a common currency. In other words, a unit of home currency (HC) should have the same purchasing power around the world. The relative version of purchasing power parity, which is used more commonly now, states that the exchange rate between the home currency and any foreign currency will adjust to reflect changes in the price levels of the two countries. For example, if inflation is 5% in the United States and 1% in Japan, then the dollar value of the Japanese yen must rise by about 4% to equalize the dollar price of goods in the two countries.

- b. *What are some reasons for deviations from purchasing power parity?*

ANSWER. PPP might not hold because:

- The price indices used to measure PPP may use different weights or different goods and services.
- Arbitrage may be too costly, because of tariffs and other trade barriers and high transportation costs, or too risky, because prices could change during the time that an item is in transit between countries.
- Since some goods and services used in the indices are not traded, there could be price discrepancies between countries.
- Relative price changes could lead to exchange rate changes even in the absence of an inflation differential.
- Government intervention could lead to a disequilibrium exchange rate.

- c. *Under what circumstances can purchasing power parity be applied?*

ANSWER. The relative version of purchasing power parity holds up best in two circumstances: (a) over long periods of time among countries with a moderate inflation differential since the general trend in the price level ratio will tend to dominate the effects of relative price changes, and (b) in the short run during periods of hyperinflation since with high inflation changes in the general level of prices quickly swamp the effects of relative price changes.

Chapter 4, Question 3

3. Suppose the dollar/rupee rate is fixed but Indonesian prices are rising faster than U.S. prices. Is the Indonesian rupee appreciating or depreciating in real terms?

ANSWER. The rupee's real value is rising since it is not depreciating to compensate for higher Indonesian inflation.

Chapter 4, Problem 2

2. Two countries, the United States and England, produce only one good, wheat. Suppose the price of wheat is \$3.25 in the United States and is £1.35 in England.

a. According to the law of one price, what should the USD/GBP spot exchange rate be?

ANSWER. Since the price of wheat must be the same in both nations, the exchange rate, e , is $3.25/1.35$ or $e = \$2.4074$.

b. Suppose the price of wheat over the next year is expected to rise to \$3.50 in the United States and to £1.60 in England. What should the one year USD/GBP forward rate be?

ANSWER. In the absence of uncertainty, the forward rate, f , should be $3.50/1.60$ or $f = \$2.1875$.

c. If the U.S. government imposes a tariff of \$0.50 per bushel on wheat imported from England, what is the maximum possible change in the spot exchange rate that could occur?

ANSWER. If e is the exchange rate, then wheat selling in England at £1.35 will sell in the United States for $1.35e + 0.5$, where 0.5 is the U.S. tariff on English wheat. In order to eliminate the possibility of arbitrage, $1.35e + 0.5$ must be greater than or equal to \$3.25, the price of wheat in the U.S. or $e > \$2.0370$. Thus the maximum exchange rate change that could occur is $(2.4074 - 2.0370)/2.4074 = 15.38\%$. This solution assumes that the pound and dollar prices of wheat remain the same as before the tariff.

Equation (4.7) from the PPT slides of chapter 4 and also from textbook by Shapiro on page 114 is reproduced below for your convenience in solving the next two problems listed. Please also note that the full explanations of Equations 4.6 and 4.7 have

$$\text{already been provided in the PPT slides for Chapter 4. } e'_t = e_t \times \frac{(1+i_f)^t}{(1+i_h)^t} \quad (4.7)$$

Chapter 4, Problem 7

7. Chase Econometrics has just published projected inflation rates for the United States and Germany for the next five years. U.S. inflation is expected to be 10 percent per year, and German inflation is expected to be 4 percent per year.

a. If the current exchange rate is \$0.95/€, what should the exchange rates for the next five years be?

ANSWER. According to PPP, the exchange rate for the euro at the end of year t should equal $0.95(1.10/1.04)^t$. Hence, projected exchange rates for the next 5 years are \$1.0048, \$1.0628, \$1.1241, \$1.1889, \$1.2575.

Equation (4.7) from the PPT slides of Ch 4 and/or the textbook by Shapiro on page 114 is reproduced below for your convenience to solve the next two problems listed below

$$e'_t = e_t \times \frac{(1+i_f)^t}{(1+i_h)^t} \quad (4.7)$$

b. Suppose that U.S. inflation over the next five years turns out to average 3.2%, German inflation averages 1.5%, and the exchange rate in five years is \$0.99/€. What has happened to the real value of the euro over this five-year period?

ANSWER. According to Equation 4.7 of the PPT slides for Chapter 4 and/or the Textbook (Shapiro), the real value of the euro at the end of five years is

$$e'_t = e_t \cdot \frac{(1+i_f)^t}{(1+i_h)^t} = 0.99 \cdot \left(\frac{1.015}{1.032}\right)^5 = 0.9111$$

Hence, even though the euro has appreciated in nominal terms over this five-year period, it has fallen in real terms by 4.09% $[(0.9111 - 0.95)/0.95]$.

Chapter 4, Problem 8

8. During 1995, the Mexican peso exchange rate rose from Mex5.33/USD to Mex7.64/USD.

At the same time, U.S. inflation was approximately 3% in contrast to Mexican inflation of about 48.7%.

a. By how much did the nominal value of the peso change during 1995?

ANSWER. During 1995, the peso fell from \$0.1876 (1/5.33) to \$0.1309 (1/7.64), which is equivalent to a devaluation of 30.24% $[(0.1309 - 0.1876)/0.1876]$

- b. *By how much did the real value of the peso change over this period?*

ANSWER. Using Equation 4.7 of the PPT slides and/or textbook (Shapiro), the real value of the peso by the end of 1995 was \$0.1890:

$$e'_t = e_t \cdot \frac{(1 + i_f)^t}{(1 + i_h)^t} = 0.1309 \cdot \frac{1.487}{1.03} = 0.1890$$

Based on this real exchange rate, the peso has appreciated during 1995 by 0.72% $((0.1890 - 0.1876)/0.1876)$. In other words, the real exchange rate stayed virtually constant, implying the purchasing power parity held during the year.

Only if time permits, you may wish to attempt these two questions.

Additional Questions

1. Assume that the price level in Canada is CAD16,600, the price level in France is EUR11,750, and the spot exchange rate is CAD 1.35/EUR.

- a. *What is the implied exchange rate of CAD/EUR that satisfies absolute PPP?*

ANSWER. The spot rate that satisfies absolute PPP is the ratio of the Canadian price level in Canadian dollars to the French price level in Euros: CAD16,600/EUR11,750 = CAD1.4128/EUR.

- b. *Is the Euro over- or undervalued relative to the Canadian Dollar?*

ANSWER. As the actual exchange rate of CAD 1.35/EUR is lower than the PPP exchange rate, the Euro is undervalued. To go back to equilibrium, it would have to increase to CAD 1.4128/EUR.

- c. *What amount of appreciation or depreciation of the Euro would be required to return the actual exchange rate to its PPP value?*

ANSWER. $(1.4128 - 1.35) / 1.35 = 0.0466 \rightarrow 4.66\%$ appreciation.

2. If the spot JPY/EUR exchange rate is 114 (i.e. 1 EUR = JPY 114), and expected inflation rates in Japan and Europe are 3% and 5%, respectively, what is the expected change in the EUR over this one year period?

ANSWER. Based on relative PPP, the expected exchange rate at $t+1 = 114 \times [(1+0.03)/(1+0.05)] = 111.83$ JPY/EUR. The % change in JPY/EUR rate is then $(111.83 - 114)/114 = -1.905\%$.

We can also calculate this directly from the relative inflation rates: $[(1+0.03)/(1+0.05)] - 1 = -0.01905 \rightarrow -1.905\%$